

NGUYEN QUANG DUNG

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Hanoi, Vietnam

SUMMARY

AI/ML Engineer with industry and research experience in physics-informed machine learning, LLM alignment, retrieval-augmented generation, and agentic systems. Builds end-to-end ML systems spanning data curation, differentiable simulation, model development, evaluation, retrieval, API serving, and deployment. Trained and deployed a production greenhouse digital twin across a 14-site fleet and delivered Web3 AI systems, while conducting research in GraphRAG, agent memory, and knowledge distillation.

EDUCATION

Hanoi University of Science and Technology
B.Sc., Computer Science

Nov. 2023 – Aug. 2027
Hanoi, Vietnam

WORK EXPERIENCE

Koidra AI

AI/ML Engineer

Nov. 2025 – Present
Remote

- Trained and shipped a physics-informed greenhouse digital twin (differentiable ODE climate simulator optimized end-to-end via BPTT) across a 14-location production fleet, reaching R^2 0.92–0.95 on air temperature and 0.67–0.82 on vapor pressure over free-running multi-day horizons.
- Built flux-level diagnostic tooling decomposing each state's heat and vapor balance per process, turning opaque prediction errors into traceable physical violations.
- Designed and validated a physics-informed transpiration model achieving an R^2 of 0.90, a 20% relative improvement over baselines.
- Integrated PID-based greenhouse control with gradient-based anomaly detection and refactored the core control module around a state-machine design, improving operational stability and reducing manual interventions in production.

Foundation Model Lab — BKAI

Undergraduate Research Assistant

Feb. 2026 – Present
Hanoi, Vietnam

- Developed a memory-augmented agent architecture integrating episodic memory, semantic retrieval, and graph-structured knowledge, achieving 74% Acc. on Locomo10 & 86% Acc. on LongMemEval-S.
- Investigated information-theoretic and model-geometry approaches for LLM knowledge distillation, focusing on transferring teacher-model knowledge through distributional signals, representation structure, and semantic alignment.

A-Star Group

AI Engineer (Promoted from Intern)

Feb. 2025 – Nov. 2025
Hanoi, Vietnam

- Led technical development of 3 Web3 AI initiatives, translating stakeholder requirements into deployable MVPs.
- Architected an MCP-based multi-agent platform that executed 10+ on-chain tools/workflows with a 95% task-completion rate.
- Delivered a FastAPI wallet-classification service trained on 3M+ wallets, achieving 75% Acc. (SOTA).
- Implemented blockchain-specific RAG and NER pipelines to improve entity grounding and reduce hallucinations in autonomous agent responses.

Natural Language Processing Lab — BKAI

Undergraduate Research Assistant

May. 2024 – Nov. 2025
Hanoi, Vietnam

- Contributed to Vietnamese legal QA research by developing data preparation, fine-tuning, and evaluation pipelines for preference-aligned language models.

RESEARCH & PUBLICATIONS

HyperProve: Answer-Guided Hypergraph Expansion for Multi-Hop Question Answering.

EMNLP 2026

An Nguyen Phu*, **Dung Nguyen Quang***, Luu Hieu An, Linh Ngo Van, Trung Le, Thien Huu Nguyen *Co-First Author*

- Developed answer-conditioned hypergraph retrieval over atomic facts, outperforming the strongest evaluated baselines by 6.2% in answer accuracy and 4.9% in F1 across three multi-hop QA benchmarks.
- **Arxiv:** HyperProve

HONORS & AWARDS

- **2nd Prize:** Vietnamese National Excellent Student Competition in Physics (VPhO)
- **Top 6 Globally:** Trustworthy NeuroSymbolic & XAI Workshop @ **IJCNN 2025**
- **Top 5 Finalist:** DataFlow 2025 (National Data Analysis Hackathon)

TECHNICAL SKILLS

- **Languages:** Python, Bash/Shell.
- **AI/ML:** PyTorch, Physics-Informed ML, Differentiable Simulation & BPTT, Time-Series Modeling, GraphRAG, Knowledge Distillation, Information Retrieval.
- **Agentic AI:** LangChain, LangGraph, Model Context Protocol, Tool Calling, Multi-Agent Systems.
- **Backend/MLOps:** FastAPI, Docker, Git, CI/CD, REST APIs, Vector Databases, MongoDB.